

Carbon and its compound

1. Carbon has atomic number

- A) 4
- B) 6
- C) 8
- D) 12

Answer: B

2. Carbon belongs to group

- A) 12
- B) 13
- C) 14
- D) 15

Answer: C

3. Valency of carbon is

- A) 1
- B) 2
- C) 3
- D) 4

Answer: D

4. Carbon forms mainly

- A) Ionic bonds
- B) Covalent bonds
- C) Metallic bonds
- D) Hydrogen bonds

Answer: B

5. The property by which carbon atoms join with each other is called

- A) Isomerism
- B) Catenation
- C) Polymerization

D) Neutralization

Answer: B

6. Carbon can form

A) Straight chains

B) Branched chains

C) Rings

D) All of these

Answer: D

7. The ability of carbon to form four bonds is called

A) Tetravalency

B) Trivalency

C) Bivalency

D) Monovalency

Answer: A

8. Organic compounds mainly contain

A) Carbon

B) Oxygen

C) Nitrogen

D) Sulphur

Answer: A

9. Methane formula

A) CH_4

B) C_2H_6

C) C_3H_8

D) C_2H_4

Answer: A

10. Ethane formula

A) C_2H_6

B) CH_4

C) C_2H_4

D) C_3H_8

Answer: A

11. Propane formula

A) C_3H_8

B) C_2H_6

C) CH_4

D) C_4H_{10}

Answer: A

12. Hydrocarbons contain

A) Carbon and hydrogen

B) Carbon and oxygen

C) Carbon and nitrogen

D) Carbon and sulphur

Answer: A

13. Saturated hydrocarbons are called

A) Alkanes

B) Alkenes

C) Alkynes

D) Alcohols

Answer: A

14. Unsaturated hydrocarbons contain

A) Double or triple bonds

B) Single bonds only

C) Ionic bonds

D) Metallic bonds

Answer: A

15. Hydrocarbon with double bond is

A) Alkane

B) Alkene

C) Alkyne

D) Alcohol

Answer: B

16. Hydrocarbon with triple bond is

A) Alkane

B) Alkene

C) Alkyne

D) Ester

Answer: C

17. General formula of alkanes

A) C_nH_{2n}

B) C_nH_{2n+2}

C) C_nH_{2n-2}

D) C_nH_n

Answer: B

18. General formula of alkenes

A) C_nH_{2n}

B) C_nH_{2n+2}

C) C_nH_{2n-2}

D) C_nH_n

Answer: A

19. General formula of alkynes

A) C_nH_{2n+2}

B) C_nH_{2n}

C) C_nH_{2n-2}

D) C_nH_n

Answer: C

20. Ethene formula

A) C_2H_4

B) C_2H_6

C) CH_4

D) C_3H_8

Answer: A

21. Ethyne formula

A) C_2H_2

B) C_2H_4

C) C_2H_6

D) CH_4

Answer: A

22. The simplest hydrocarbon is

A) Methane

B) Ethane

C) Propane

D) Butane

Answer: A

23. Compounds with same molecular formula but different structure are

A) Isomers

B) Polymers

C) Radicals

D) Ions

Answer: A

24. The phenomenon is called

A) Isomerism

B) Neutralization

C) Polymerization

D) Reduction

Answer: A

25. Ethanol formula

A) C_2H_5OH

B) CH_3COOH

C) C_2H_4

D) C_2H_2

Answer: A

26. Ethanoic acid formula

A) CH_3COOH

B) $\text{C}_2\text{H}_5\text{OH}$

C) C_2H_6

D) CH_4

Answer: A

27. Alcohol functional group

A) $-\text{OH}$

B) $-\text{COOH}$

C) $-\text{CHO}$

D) $-\text{CO}$

Answer: A

28. Carboxylic acid functional group

A) $-\text{OH}$

B) $-\text{COOH}$

C) $-\text{CHO}$

D) $-\text{NH}_2$

Answer: B

29. Ethanoic acid is commonly called

A) Vinegar

B) Alcohol

C) Petrol

D) Ether

Answer: A

30. Reaction between acid and alcohol forming ester is

A) Esterification

B) Neutralization

C) Oxidation

D) Reduction

Answer: A

31. Esterification produces

A) Ester and water

B) Acid and water

C) Alcohol and water

D) Salt and water

Answer: A

32. Esterification requires

A) Acid catalyst

B) Base catalyst

C) Salt catalyst

D) Metal catalyst

Answer: A

33. Hydrolysis of ester produces

A) Acid and alcohol

B) Salt and water

C) Acid and salt

D) Alcohol and salt

Answer: A

34. Combustion of hydrocarbons produces

A) CO_2 and H_2O

B) CO and H_2O

C) CO_2 only

D) H_2 only

Answer: A

35. Incomplete combustion produces

A) Carbon monoxide

B) Carbon dioxide

C) Oxygen

D) Nitrogen

Answer: A

36. Carbon monoxide is

A) Poisonous gas

B) Safe gas

C) Noble gas

D) Inert gas

Answer: A

37. Diamond is an allotrope of

A) Carbon

B) Oxygen

C) Nitrogen

D) Hydrogen

Answer: A

38. Graphite is an allotrope of

A) Carbon

B) Oxygen

C) Nitrogen

D) Hydrogen

Answer: A

39. Diamond is

A) Hardest substance

B) Soft metal

C) Liquid

D) Gas

Answer: A

40. Graphite is used as

A) Lubricant

B) Fuel

C) Acid

D) Base

Answer: A

41. Graphite conducts electricity because

A) Free electrons

B) Protons

C) Neutrons

D) Atoms

Answer: A

42. Diamond structure is

A) Tetrahedral

B) Linear

C) Planar

D) Cubic

Answer: A

43. Carbon dioxide formula

A) CO_2

B) CO

C) C_2O

D) C_2O_2

Answer: A

44. Carbon monoxide formula

A) CO

B) CO_2

C) C_2O

D) C_2O_2

Answer: A

45. Carbon dioxide is used in

A) Fire extinguishers

B) Fuel

C) Metals

D) Plastic

Answer: A

46. Carbon compounds are studied in

A) Organic chemistry

B) Inorganic chemistry

C) Physical chemistry

D) Analytical chemistry

Answer: A

47. LPG mainly contains

A) Propane and butane

B) Methane

C) Ethane

D) Ethene

Answer: A

48. CNG mainly contains

A) Methane

B) Ethane

C) Propane

D) Butane

Answer: A

49. Petrol is mixture of

A) Hydrocarbons

B) Alcohols

C) Acids

D) Salts

Answer: A

50. Kerosene is obtained from

A) Petroleum

B) Coal

C) Wood

D) Gas

Answer: A

51. Carbon chains are called

A) Carbon skeleton

B) Metal chain

C) Salt chain

D) Acid chain

Answer: A

52. Homologous series have

A) Similar chemical properties

B) Same molecular mass

C) Same structure

D) Same density

Answer: A

53. Successive members differ by

A) CH_2 group

B) CH_3 group

C) C_2H_2 group

D) H_2 group

Answer: A

54. Methanol formula

A) CH_3OH

B) $\text{C}_2\text{H}_5\text{OH}$

C) CH_4

D) CO_2

Answer: A

55. Ethanol is used as

A) Fuel

B) Solvent

C) Medicine

D) All of these

Answer: D

56. Soap is formed from

A) Fats and oils

B) Metals

C) Salts

D) Acids

Answer: A

57. Soap is produced by

A) Saponification

B) Esterification

C) Neutralization

D) Combustion

Answer: A

58. Detergents clean even in

A) Hard water

B) Soft water

C) Distilled water

D) Rainwater

Answer: A

59. Hard water contains

A) Calcium and magnesium salts

B) Sodium salts

C) Potassium salts

D) Chlorine salts

Answer: A

60. Soap does not lather in

A) Hard water

B) Soft water

C) Distilled water

D) Rainwater

Answer: A

61. Micelles are formed by

A) Soap molecules

B) Salt molecules

C) Acid molecules

D) Metal molecules

Answer: A

62. Soap molecules have

A) Hydrophilic and hydrophobic ends

B) Only hydrophilic end

C) Only hydrophobic end

D) No ends

Answer: A

63. Hydrophilic means

A) Water loving

B) Water hating

C) Oil loving

D) Oil hating

Answer: A

64. Hydrophobic means

A) Water hating

B) Water loving

C) Acid loving

D) Base loving

Answer: A

65. Carbon compounds burn in

A) Oxygen

B) Nitrogen

C) Hydrogen

D) Helium

Answer: A

66. Combustion releases

A) Heat and light

B) Only heat

C) Only light

D) Only gas

Answer: A

67. Ethanol burns forming

A) CO_2 and H_2O

B) CO

C) H_2

D) O_2

Answer: A

68. Carbon compounds are mostly

A) Covalent

B) Ionic

C) Metallic

D) Nuclear

Answer: A

69. Covalent compounds have

A) Low melting point

B) High melting point

C) Very high density

D) High conductivity

Answer: A

70. Covalent compounds are

A) Poor conductors

B) Good conductors

C) Metals

D) Ionic

Answer: A

71. Structural formula shows

A) Arrangement of atoms

B) Mass

C) Density

D) Temperature

Answer: A

72. Molecular formula shows

A) Number of atoms

B) Arrangement

C) Mass only

D) Charge

Answer: A

73. Diamond is used in

A) Cutting tools

B) Fuel

C) Fertilizer

D) Medicine

Answer: A

74. Graphite is used in

A) Pencil lead

B) Fuel

C) Salt

D) Acid

Answer: A

75. Fullerene is

A) Allotrope of carbon

B) Metal

C) Gas

D) Acid

Answer: A

76. Buckminster fullerene formula

A) C₆₀

B) C₂₀

C) C₄₀

D) C₁₀

Answer: A

77. Carbon forms strong bonds due to

A) Small atomic size

B) Large atomic size

C) High mass

D) High density

Answer: A

78. Organic compounds are mostly

A) Covalent

B) Ionic

C) Metallic

D) Nuclear

Answer: A

79. Ethanoic acid smells like

A) Vinegar

B) Petrol

C) Alcohol

D) Gas

Answer: A

80. Ester smells

A) Fruity

B) Sour

C) Bitter

D) Odourless

Answer: A

81. Carbon compounds are used in

A) Fuels

B) Plastics

C) Medicines

D) All of these

Answer: D

82. Plastics are

A) Polymers

B) Metals

C) Salts

D) Acids

Answer: A

83. Polymer is

A) Large molecule

B) Small atom

C) Metal

D) Salt

Answer: A

84. Polymerization forms

A) Polymers

B) Metals

C) Acids

D) Salts

Answer: A

85. Polyethylene is used for

A) Plastic bags

B) Fuel

C) Fertilizer

D) Acid

Answer: A

86. Carbon forms compounds with

A) Hydrogen

B) Oxygen

C) Nitrogen

D) All of these

Answer: D

87. Organic chemistry studies

A) Carbon compounds

B) Metals

C) Salts

D) Gases

Answer: A

88. Hydrocarbons are main components of

A) Petroleum

B) Salt

C) Water

D) Air

Answer: A

89. Petroleum products include

A) Petrol

B) Diesel

C) Kerosene

D) All of these

Answer: D

90. Carbon dioxide turns lime water

A) Milky

B) Blue

C) Red

D) Green

Answer: A

91. Lime water test detects

A) CO₂

B) O₂

C) H₂

D) N₂

Answer: A

92. Carbon monoxide binds with

A) Haemoglobin

B) Oxygen

C) Nitrogen

D) Water

Answer: A

93. CO poisoning prevents

A) Oxygen transport

B) Digestion

C) Breathing

D) Drinking

Answer: A

94. Methane is

A) Greenhouse gas

B) Noble gas

C) Metal

D) Salt

Answer: A

95. Greenhouse gases cause

A) Global warming

B) Cooling

C) Freezing

D) Rain

Answer: A

96. Carbon cycle maintains

A) Balance of carbon

B) Balance of oxygen

C) Balance of nitrogen

D) Balance of metals

Answer: A

97. Photosynthesis uses

A) CO_2

B) O_2

C) N_2

D) H_2

Answer: A

98. Plants convert CO_2 into

A) Glucose

B) Salt

C) Metal

D) Acid

Answer: A

99. Respiration releases

A) CO_2

B) O_2

C) N_2

D) H_2

Answer: A

100. Carbon compounds are essential in

A) Life processes

B) Industry

C) Daily life

D) All of these

Answer: D